



STRATHMORE UNIVERSITY
Institute of Mathematical Sciences

2025/2026
1ST YEAR, SEMIMESTER III

COURSE OUTLINE:

DSA 8301: Statistical Inference for Big Data

STATEMENT ABOUT ACADEMIC HONESTY AND INTEGRITY

Upon registering as bonafide students at Strathmore University, all students pledge to adhere to the institution's policies and regulations. Upholding academic honesty and integrity is of utmost importance to guarantee the validity, reliability, and credibility of the learning process.

Strathmore University unequivocally opposes any type of academic dishonesty, encompassing actions such as plagiarism, cheating, and engaging in other deceptive practices during assessments. The university insists that students consistently complete their own assignments and generate their academic work, except in cases where group assignments are explicitly assigned. Any manifestation of academic dishonesty is considered a breach of conduct according to Strathmore University's Rules and Regulations.

COURSE INFORMATION

COURSE CODE : DSA 8301:

COURSE TITLE: Statistical Inference for Big Data

CREDIT HOURS: 4 hours per week

COURSE PURPOSE:

The aim of this module is to give a solid foundation to the understanding of statistics as a general approach to the problem of making valid inferences about relationships using data from observational or experimental studies including classical techniques of hypothesis testing and point and interval estimation, and nonparametric methods.

LEARNING OUTCOMES

On completing the course students will, through assessment activities, show evidence of their ability to:

1. Describe the principal features of, and differences between, frequentist, likelihood and Bayesian inference;
2. Define and derive the likelihood function based on data from a parametric statistical model, and describe its role in various forms of inference;
3. Define, derive and apply different methods for evaluating and comparing estimators;
4. Describe, derive and apply lower bounds for the variance of an unbiased estimator;

5. Define, derive and apply the error probabilities of a test between two simple hypotheses; define and conduct a likelihood ratio test; state and apply the Neyman-Pearson lemma;

COURSE DESCRIPTION:

Parametric statistical models; statistical estimation and sampling distributions; properties of estimators (bias, consistency, efficiency, sufficiency, robustness); asymptotic properties of estimators; maximum likelihood estimation of model parameters, asymptotic distributions of the maximum likelihood estimator. Statistical hypothesis testing, Type I and II errors, power, likelihood ratios; properties of statistical tests, application of likelihood inference to simple statistical analyses including linear regression and contingency tables. Large sample theory, including convergence concepts, and asymptotic concepts in inference including finite sample and asymptotic efficiency of inference procedures. Nonparametric statistical methods: overview of nonparametric inference, exact p-values for tests, exact coverage probabilities for confidence intervals, exact experiment wise error rates for multiple comparison procedures, and exact coverage probabilities for confidence bands. Dichotomous Data: One-Sample and Two-Sample Location Problems. Point Estimation and Confidence Intervals. Asymptotic relative comparisons. Density estimation. Confidence sets. Computer exercises using real world data.

PRE-REQUISITES:

DSA 8405:

MODE OF DELIVERY:

The following methods of delivery will be used in this course: lectures, seminars, practical data analysis sessions, consulting workshops/clinics, group discussion, and case studies

EFFECTIVE DATE: 28th November 2025

LECTURER INFORMATION:

Lecturer's name	Prof Jacob Ong'ala
Email	jongala@strathmore.edu
Phone	0710945685

Course Resources:

Core Reading Materials with Active Links

1. Casella, G., & Berger, R. L. (2002). *Statistical Inference* (2nd ed.). Duxbury / Cengage Learning.
PDF available (online lecture copy): https://pages.stat.wisc.edu/~shao/stat610/Casella_Berger_Statistical_Inference.pdf :contentReference[oaicite:1]index=1
2. Lehmann, E. L., & Romano, J. P. (1986). *Testing Statistical Hypotheses*. Wiley.
Older freely browsable edition: <https://gwern.net/doc/statistics/decision/1986-lehmann-testings.pdf> :contentReference[oaicite:2]index=2
3. Archive.org free copy (older publishing) of Casella & Berger: <https://archive.org/details/statisticalinfer0000case> :contentReference[oaicite:3]index=3

Recommended Reading Materials with Active Links

1. Gibbons, J. D., & Chakraborti, S. (2011). *Nonparametric Statistical Inference*. CRC Press.
Free preview excerpts and exercise scans are often available via internet archives and course

PDF repositories (search by title + “pdf”).

2. Silverman, B. W. (1986). *Density Estimation for Statistics and Data Analysis*. Chapman & Hall.

Select preview materials sometimes available via open PDFs in university course sites (searchable online).

COURSE SCHEDULE

Topic 1: Parametric Statistical Models

- Definition and formulation of parametric statistical models.
- Common families of parametric distributions.
- Model assumptions and identifiability issues.

Topic 2: Statistical Estimation and Sampling Distributions

- Concept of point estimation.
- Sampling distributions of estimators.
- Method of moments and basic estimation techniques.

Topic 3: Properties of Estimators

- Bias and unbiasedness of estimators.
- Consistency of estimators.
- Efficiency and sufficiency.
- Robustness considerations.

Topic 4: Asymptotic Properties of Estimators

- Modes of convergence.
- Law of Large Numbers.
- Central Limit Theorem.
- Asymptotic normality and efficiency.

Topic 5: Maximum Likelihood Estimation

- Likelihood and log-likelihood functions.
- Derivation of maximum likelihood estimators.
- Fisher information.
- Asymptotic distribution of the MLE.

Topic 6: Statistical Hypothesis Testing

- Formulation of null and alternative hypotheses.
- Type I and Type II errors.
- Power of statistical tests.
- Likelihood ratio tests and their properties.

Topic 7: Likelihood Inference and Applications

- Likelihood-based confidence intervals.
- Inference in linear regression models.
- Likelihood inference for contingency tables.

Topic 8: Large Sample Theory

- Finite-sample versus asymptotic inference.
- Asymptotic concepts in estimation and testing.
- Efficiency of inference procedures in large samples.

Topic 9: Nonparametric Statistical Methods

- Overview of nonparametric inference.
- Exact tests and exact p-values.
- Exact confidence intervals and confidence bands.
- Multiple comparison procedures and experiment-wise error rates.

Topic 10: Specialized Topics and Computational Applications

- Dichotomous data: one-sample and two-sample location problems.
- Density estimation techniques.
- Confidence sets and asymptotic relative efficiency.

ASSESSMENT AND EVALUATION STRATEGIES:

Assignments		Weight
1	Course Work Assessment	40%
2	End of Semester Written Examination	60%

DATE REVISED: 2nd Feb 2026

AUTHORISATION:

This course is authorized for use by:

Prof. Jacob Ong'ala
Lecturer

Signature of Lecturer

Date